Universidad del Desarrollo
Facultad de Ingeniería

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Listado 7 - Álgebra

Polinomios II: Fracciones Parciales

1. $\dfrac{11}{x^2+3x-28}$

2. $\dfrac{-x+22}{x^2-2x-8}$

3. $\dfrac{3x-13}{6x^2-x-12}$

4. $\dfrac{x^2-12x+18}{x^3-6x^2+9x}$

5. $\dfrac{5x^2+3x+6}{x^3+2x^2+3x}$

6. $\dfrac{2x^3+7x+5}{x^4+4x^2+4}$

7. $\dfrac{x^3-7x^2+17x-17}{x^2-5x+6}$

8. $\dfrac{4x^2+5x-9}{x^3-6x-9}$

9. $\dfrac{4x^2-8x+1}{x^3-x+6}$

10. $\dfrac{11x-11}{6x^2+7x-3}$

11. $\dfrac{3x^2+x}{(x-2)(x^2+3)}$

12. $\dfrac{17x-1}{(2x-3)(3x-1)}$

13. $\dfrac{6x^2-15x+16}{x^3-3x^2+4x}$

14. $\dfrac{-5x^2+7x-18}{x^4+6x^2+9}$

15. $\dfrac{x^3+x^2-13x+11}{x^2+2x-15}$

1. $\dfrac{11}{x^2+3x-28}$

$$\frac{11}{x^2+3x-28} = \frac{11}{(x+7)(x-4)} = \frac{A}{x+7} + \frac{B}{x-4}$$

$$11 = A(x-4) + B(x+7)$$

o Si $x = 4$ o Si $x = -7$

$11 = 11B$ $11 = -11A$

$B = 1$ $A = -1$

$$\frac{11}{x^2+3x-28} = \frac{-1}{x+7} + \frac{1}{x-4}$$

2. $\dfrac{-x+22}{x^2-2x-8}$

$$\frac{-x+22}{x^2-2x-8} = \frac{-x+22}{(x-4)(x+2)} = \frac{A}{x-4} + \frac{B}{x+2}$$

$$-x+22 = A(x+2) + B(x-4)$$

o Si $x = -2$,   o Si $x = 4$,

$24 = -6B$ $18 = 6A$

$B = -4$ $A = 3$

$$\frac{-x+22}{x^2-2x-8} = \frac{3}{x-4} - \frac{4}{x+2}$$

3. $\dfrac{3x-13}{6x^2-x-12}$

$$\frac{3x-13}{6x^2-x-12} = \frac{3x-13}{\left(x-\frac{3}{2}\right)\left(x+\frac{4}{3}\right)} = \frac{A}{x-\frac{3}{2}} + \frac{B}{x+4/3}$$

$$3x-13 = A\left(x+4/3\right) + B\left(x-\tfrac{3}{2}\right)$$

o si $x = -4/3$ o si $x = \frac{3}{2}$

$$-17 = \frac{-17B}{3} \qquad \frac{-17}{2} = \frac{17}{6}A$$

$$B = 3 \qquad A = -3$$

$$\frac{3x-13}{6x^2-x-12} = -\frac{3}{x-\frac{3}{2}} + \frac{3}{x+4/3}$$

4. $\dfrac{x^2-12x+18}{x^3-6x^2+9x}$

$$\frac{x^2-12x+18}{x^3-6x^2+9x} = \frac{x^2-12x+18}{x(x-3)^2} = \frac{A}{x} + \frac{B}{x-3} + \frac{C}{(x-3)^2}$$

$$x^2-12x+18 = A(x-3)^2 + B(x-3)x + Cx$$
$$x^2-12x+18 = Ax^2-6Ax+9A+Bx^2-3Bx+Cx$$
$$x^2-12x+18 = (A+B)x^2 + (-6A-3B+C)x + 9A$$

$A+B = 1 \Rightarrow B = -1$

$-6A-3B+C = -12 \Rightarrow -12+3+C = -12 \Rightarrow C = -3$

$9A = 18 \Rightarrow A = 2$

$$\frac{x^2-12x+18}{x^3-6x^2+9x} = \frac{2}{x} - \frac{1}{x-3} - \frac{3}{(x-3)^2}$$

5. $$\frac{5x^2+3x+6}{x^3+2x^2+3x}$$

$$\frac{5x^2+3x+6}{x(x^2+2x+3)} = \frac{A}{x} + \frac{Bx+C}{x^2+2x+3}$$

$$5x^2+3x+6 = A(x^2+2x+3)+(Bx+C)x$$

$$5x^2+3x+6 = Ax^2+2Ax+3A+Bx^2+Cx$$

$$5x^2+3x+6 = (A+B)x^2+(A+C)x+3A$$

$A+B=5 \Rightarrow B=3$

$A+C=3 \Rightarrow C=1$ $\Rightarrow \dfrac{5x^2+3x+6}{x^3+2x^2+3x} = \dfrac{2}{x} + \dfrac{3x+1}{x^2+2x+3}$

$3A=6 \Rightarrow A=2$

6. $$\frac{2x^3+7x+5}{x^4+4x^2+4}$$

$$\frac{2x^3+7x+5}{(x^2+2)^2} = \frac{Ax+B}{x^2+2} + \frac{Cx+D}{(x^2+2)^2}$$

$$2x^3+7x+5 = (Ax+B)(x^2+2)+(Cx+D)$$

$$2x^3+7x+5 = Ax^3+2Ax+Bx^2+2B+Cx+D$$

$$2x^3+7x+5 = Ax^3+Bx^2+(2A+C)x+D+2B$$

$Ax^3 = 2x^3 \Rightarrow A=2$

$Bx^2 = 0 \Rightarrow B=0$

$2A+C=7 \Rightarrow C=3$ $\Rightarrow \dfrac{2x}{x^2+2} + \dfrac{3x+5}{(x^2+2)^2}$

$D+2B=5 \Rightarrow D=5$

7. $\dfrac{x^3-7x^2+17x-17}{x^2-5x+6}$

1º) Dividimos

$$x^3-7x^2+17x-17 : x^2-5x+6 = x-2$$

$$-(x^3-5x^2+6x)$$

$$-2x^2+11x-17$$

$$-(-2x^2+10x-12)$$

$$-21x-5$$

$$x-2 - \frac{21x+5}{x^2-5x+6}$$

$$\frac{-21x-5}{(x-3)(x-2)} \Rightarrow \frac{A}{x-3} + \frac{B}{x-2}$$

$$-21x-5 = A(x-2)+B(x-3)$$

o Si $x=2$, • Si $x=3$

$$-47 = -B \qquad -68 = A$$

$$B = 47 \qquad A = -68$$

$$\frac{x^3-7x^2+17x-17}{x^2-5x+6} = x-2 - \frac{68}{x-3} + \frac{47}{x-2}$$

8. $$\frac{4x^2+5x-9}{x^3-6x-9}$$

$$\frac{4x^2+5x-9}{(x-3)(x^2+3x+3)} = \frac{A}{x-3} + \frac{Bx+C}{x^2+3x+3}$$

$$4x^2+5x-9 = Ax^2+3Ax+3A+Bx^2-3Bx+Cx-3C$$

$$4x^2+5x-9 = (A+B)x^2+(3A+3B+C)x+3A-3C$$

$$A(x^2+3x+3)+(Bx+C)(x-3)$$

Si $x = 3 \Rightarrow 42 = 21A + 0$

$A = 2$

$A + B = 4 \Rightarrow B = 2$

$3A + 3B + C = 5 \Rightarrow 6 + 6 + C = 5 \Rightarrow C = -7$

$3A - 3C = -9 \Rightarrow$

$$\frac{4x^2+5x-9}{x^3-6x-9} = \frac{2}{x-3} + \frac{2x-7}{x^2+3x+3}$$

9. $\dfrac{4x^2 - 8x + 1}{x^3 - x + 6}$

$$\frac{4x^2 - 8x + 1}{(x+2)(x^2 - 2x + 3)}$$

$$\frac{A}{x+2} + \frac{Bx + C}{x^2 - 2x + 3}$$

$$4x^2 - 8x + 1 = A(x^2 - 2x + 3) + (Bx + C)(x + 2)$$

$$Ax^2 - 2Ax + 3A + Bx^2 + 2Bx + Cx + 2C$$

$$(A + B)x^2 + (-2A + 2B + C)x + 3A + 2C$$

Si $x = -2$

$$33 = 11A$$

$$A = 3$$

$A + B = 4 \Rightarrow B = 1$

$-2A + 2B + C = -8$

$3A + 2C = 1 \Rightarrow C = -4$

$$\frac{4x^2 - 8x + 1}{x^3 - x + 6} = \frac{3}{x+2} + \frac{x - 4}{x^2 - 2x + 3}$$

10. $$\frac{11x-11}{6x^2+7x-3}$$

$$\frac{11x-11}{(3x-1)(2x+3)} \Rightarrow \frac{A}{3x-1} + \frac{B}{2x+3}$$

$$11x - 11 = A(2x+3) + B(3x-1)$$

- Si $x = -\frac{3}{2}$

$$-\frac{33}{2} - 11 = -\frac{11B}{2}$$

$$-\frac{55}{2} = -\frac{11}{2}B$$

$$5 = B$$

- Si $x = \frac{1}{3}$

$$\frac{11}{3} - 11 = \frac{11A}{3}$$

$$-\frac{22}{3} = \frac{11}{3}A$$

$$A = -2$$

$$\frac{11x-11}{6x^2+7x-3} = -\frac{2}{3x-1} + \frac{5}{2x+3}$$

11. $$\frac{3x^2+x}{(x-2)(x^2+3)}$$

$$\frac{3x^2+3}{(x-2)(x^2+3)} \Rightarrow \frac{A}{x-2} + \frac{Bx+C}{x^2+3}$$

$$3x^2 + x = A(x^2+3) + (Bx+C)(x-2)$$

$$3x^2 + x = Ax^2 + 3A + Bx^2 - 2Bx + Cx - 2C$$

$$3x^2 + x = (A+B)x^2 + (-2B+C)x + 3A - 2C$$

- Si $x = 2$

$$14 = 7A$$

$$2 = A$$

- $A + B = 3 \Rightarrow B = 1$
- $-2B + C = 1 \Rightarrow C = 3$
- $3A - 2C = 0$

$$\frac{3x^2+x}{(x-2)(x^2+3)} = \frac{2}{x-2} + \frac{x+3}{x^2+3}$$

12. $\dfrac{17x-1}{(2x-3)(3x-1)}$

$$\frac{17x-1}{(2x-3)(3x-1)} = \frac{A}{2x-3} + \frac{B}{3x-1}$$

$$17x-1 = A(3x-1) + B(2x-3)$$

o si $x = \frac{1}{3}$

$$\frac{14}{3} = -\frac{7}{3}B$$

$B = -2$

o si $x = \frac{3}{2}$

$$\frac{49}{2} = \frac{7}{2}A$$

$A = 7$

$$\frac{17x-1}{(2x-3)(3x-1)} = \frac{7}{2x-3} - \frac{2}{3x-1}$$

13. $\dfrac{6x^2-15x+16}{x^3-3x^2+4x}$

$$\frac{6x^2-15x-16}{x(x^2-3x+4)} = \frac{A}{x} + \frac{Bx+C}{x^2-3x+4}$$

$$6x^2-15x-16 = Ax^2-3Ax+4A+Bx^2+Cx$$

$$6x^2-15x-16 = (A+B)x^2+(-3A+C)x+4A$$

$$\left.\begin{array}{l} \text{o } A+B=6 \Rightarrow B=10 \\ -3A+C=-15 \Rightarrow C=-27 \\ 4A=-16 \Rightarrow A=-4 \end{array}\right\} \quad \frac{6x^2-15x-16}{x(x^2-3x+4)} = -\frac{4}{x} + \frac{10x-27}{x^2-3x+4}$$

14. $\dfrac{-5x^2+7x-18}{x^4+6x^2+9}$

$$\frac{-5x^2+7x-18}{(x^2+3)^2} = \frac{Ax+B}{x^2+3} + \frac{Cx+D}{(x^2+3)^2}$$

$$-5x^2+7x-18 = (Ax+B)(x^2+3) + Cx+D$$

$$-5x^2+7x-18 = Ax^3+3Ax+Bx^2+3B+Cx+D$$

$$-5x^2+7x-18 = Ax^3+Bx^2+(3A+C)x+3B+D$$

$$\left.\begin{array}{ll} Ax^3 = 0 & \Rightarrow A=0 \\ Bx^2 = -5x^2 & \Rightarrow B=-5 \\ 3A+C=7 & \Rightarrow C=7 \\ 3B-D=-18 & \Rightarrow D=3 \end{array}\right\} \frac{-5x^2+7x-18}{(x^2+3)^2} = -\frac{5}{x^2+3} + \frac{7x+3}{(x^2+3)^2}$$

15. $$\frac{x^3+x^2-13x+11}{x^2+2x-15}$$

$$x^3 + x^2 - 13x + 11 \ : \ x^2 + 2x - 15 = x - 1$$

$$-(x^3 + 2x^2 - 15x)$$

$$-x^2 + 2x + 11$$

$$-(-x^2 - 2x + 15)$$

$$4x - 4$$

$$x - 1 + \frac{4x-4}{x^2+2x-15} \rightarrow (x+5)(x-3)$$

$$\frac{4x-4}{x^2+2x-15} = \frac{A}{x+5} + \frac{B}{x-3}$$

$$4x - 4 = A(x-3) + B(x+5)$$

- Si $x = 3$
  $8 = 8B$
  $B = 1$
- Si $x = -5$
  $-24 = -8A$
  $3 = A$

$$\frac{4x-4}{x^2+2x-15} = \frac{3}{x+5} + \frac{1}{x-3}$$

P1) Det. Todas las raices de $2x^4+4x^3+3x^2+ax+b$ si se sabe que $1+i$ es raiz

Sabemos que $1-i$ tambien es raiz

$$\begin{array}{ccccc|l} 2 & 4 & 3 & a & b & \\ \hline & 2+2i & 4+8i & -1+15i & -16+a+(a+14)i & c=1+i \\ 2 & 6+2i & 7+8i & a-1+15i & -16+b+a+(a+14)i & \end{array}$$

$$\begin{array}{cccc|l} 2 & 6+2i & 7+8i & a-1+15i & \\ \hline & 2-2i & 8-8i & 15-15i & c=1-i \\ 2 & 8 & 15 & a+14 & \end{array}$$

Para ser raiz $a+14=0$ $\longrightarrow$ $-16+b-14 \cancel{+(-14+14)i}$

$a=-14$ $\qquad b=30$

$2x^2+8x+15$

$$\frac{-8\pm\sqrt{64-4\cdot 2\cdot 15}}{4} \Rightarrow \frac{-8\pm\sqrt{-56}}{4}$$

raices: $(1+i)(1-i)\left(\frac{-8+\sqrt{-56}}{4}\right)\left(\frac{-8-\sqrt{-56}}{4}\right)$

P2) Encontrar raices :

1) $3x^3 - 26x^2 + 52x - 24$

Pos : $\pm\frac{1}{1}, \pm\frac{1}{3}, \pm\frac{2}{1}, \pm\frac{2}{3}, \pm\frac{4}{1}, \pm\frac{4}{3}, \pm\frac{6}{1}, \pm\frac{6}{3}, \pm\frac{8}{1}, \pm\frac{8}{3}, \pm\frac{12}{1}, \pm\frac{12}{3}, \pm\frac{24}{1}, \pm\frac{24}{3}$

3	-26	52	-24	
	18	48	24	c = 6
3	-8	4	0	
	6	4		c = 2
3	-2	0		

$\Rightarrow$ $x-6$ factor, 6 raiz $\big\}$ $(3x^2-8x+4)(x-6)$

$\Rightarrow$ $x-2$ factor, 2 raiz $\big\}$ $(3x-2)(x-2)(x-6)$

raices: $\frac{2}{3}, 6, 2$

2) $4x^3 + 20x^2 - 23x + 6$

Pos: $\pm\frac{1}{1}, \pm\frac{2}{1}, \pm\frac{3}{1}, \pm\frac{6}{1}, \pm\frac{1}{2}, \pm\overset{1}{\cancel{\frac{2}{2}}}, \pm\frac{3}{2}, \pm\overset{3}{\cancel{\frac{6}{2}}}; \pm\frac{1}{4}, \pm\overset{1/2}{\cancel{\frac{2}{4}}}, \pm\frac{3}{4}, \pm\overset{3/2}{\cancel{\frac{6}{4}}}$

4	20	-23	6	
	2	11	-6	c = 1/2
4	22	-12	0	
	2	12		c = 1/2
4	24	0		

$\Rightarrow$ $x-\frac{1}{2}$ es factor, $\frac{1}{2}$ raiz $\big\}$ $(4x^2+22x-12)(x-\frac{1}{2})$

$2x^2+11x-6$

$x-1$ factor, $\frac{1}{2}$ raiz doble $\big\}$ $(4x+24)(x-\frac{1}{2})^2$

raices: $\frac{1}{2}, -5$